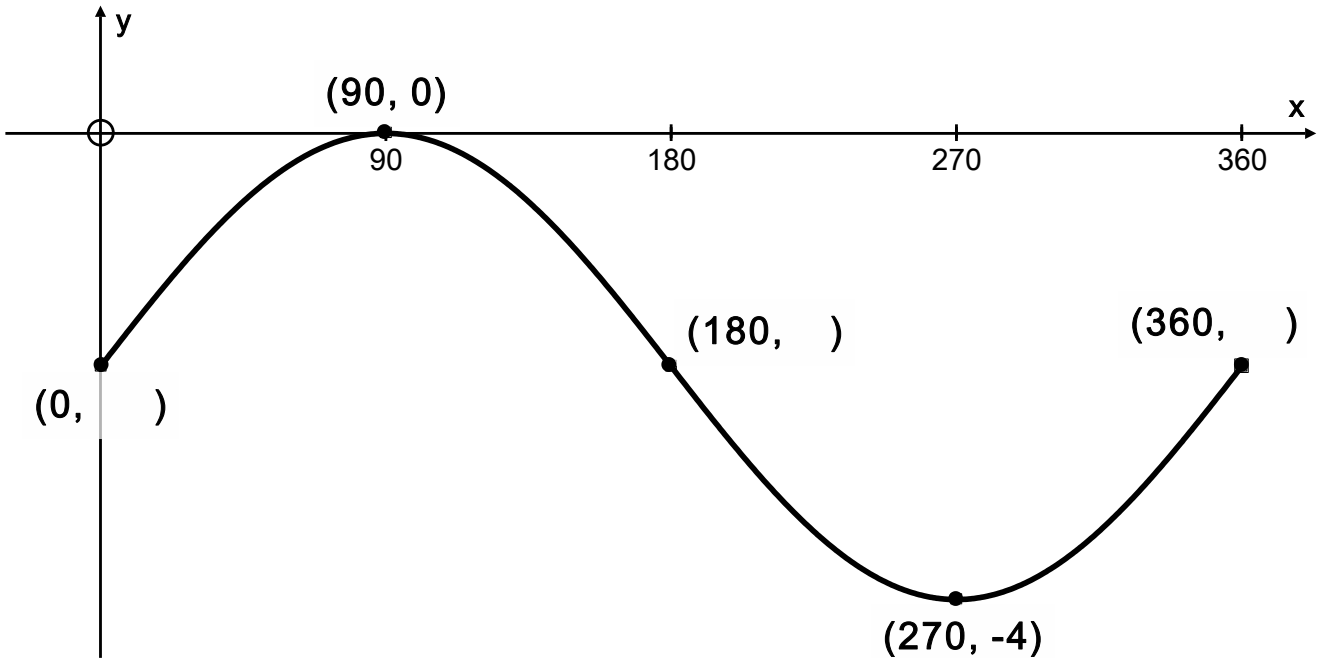
TRIG FUNCTIONS CONSOLIDATION

FINDING THE EQUATION OF TRIG GRAPHS

$$y = a\sin x + q$$

Example:

Sketched $f(x) = a \sin x + q$



Determine the values of *a* and *q*.

Working:

To find a: $|a| = \dfrac{Max\ value - Min\ value}{2} = \dfrac{0-(-4)}{2} = 2$

The sign of a : The graph is going up (increasing) from the y-intercept, $\therefore a > 0$

$\therefore a = 2$

To find q: $q = \dfrac{Max\ value + Min\ value}{2} = \dfrac{0+(-4)}{2} = \dfrac{-4}{2} = -2$

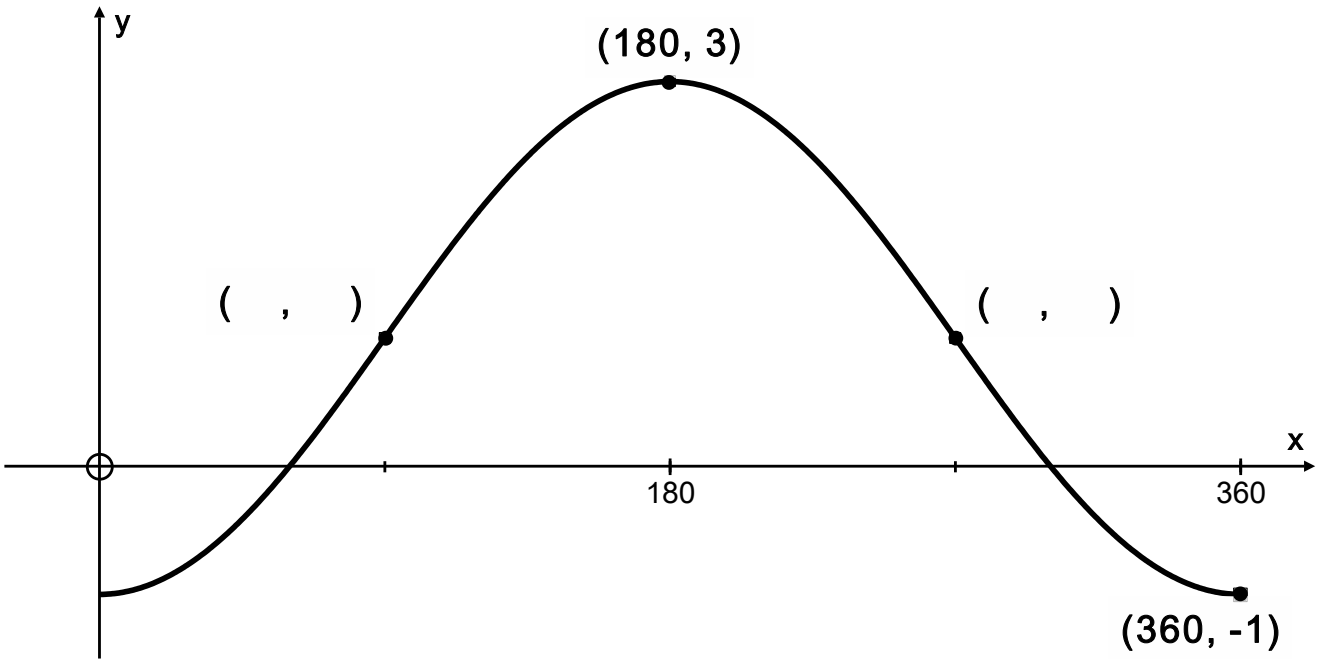
$\therefore q = -2$

$f(x) = 2\sin x - 2$

$$y = a\cos x + q$$

Example:

Sketched $f(x) = a\cos x + q$



Determine the values of *a* and *q*.

Working:

To find a: $|a| = \dfrac{Max\ value - Min\ value}{2} = \dfrac{3-(-1)}{2} = 2$

The sign of a : The graph is going up (increasing) from the y-intercept, $\therefore a < 0$

$\therefore a = -2$

To find q: $q = \dfrac{Max\ value + Min\ value}{2} = \dfrac{3+(-1)}{2} = 1$

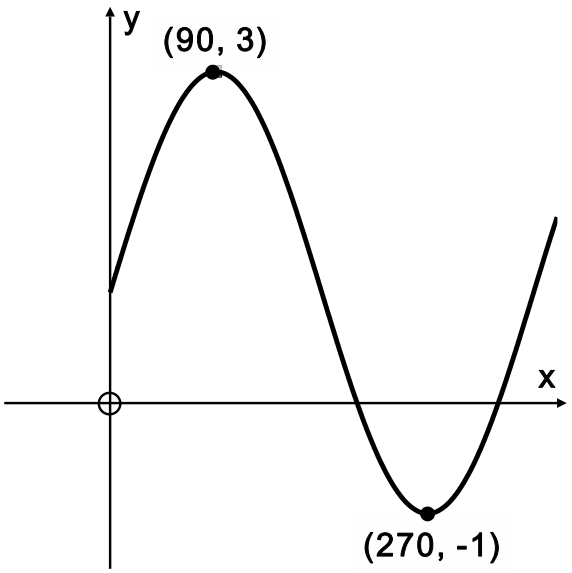
$\therefore q = 1$

$f(x) = -2\cos x + 1$

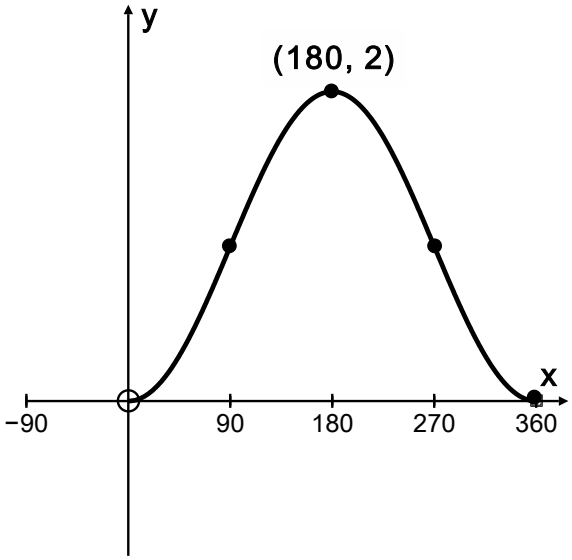
Exercise A:

Find the values of *a* and *q* in each of the following:

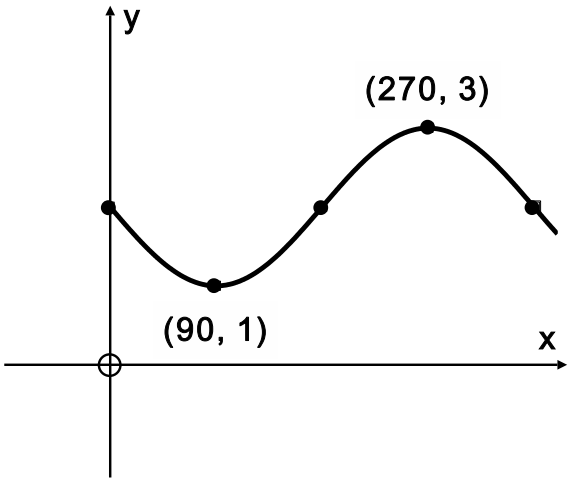
1. $y = a\sin x + q$



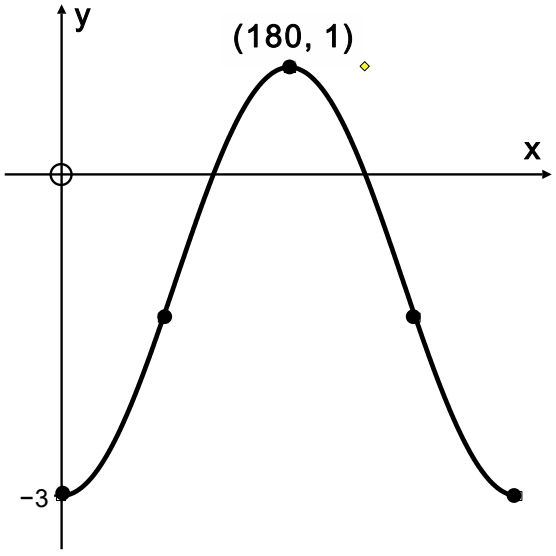
2. $y = a\cos x + q$



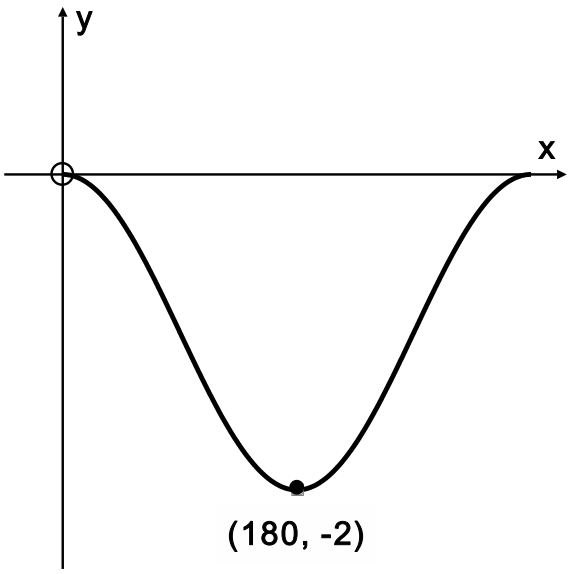
3. $y = a\sin x + q$



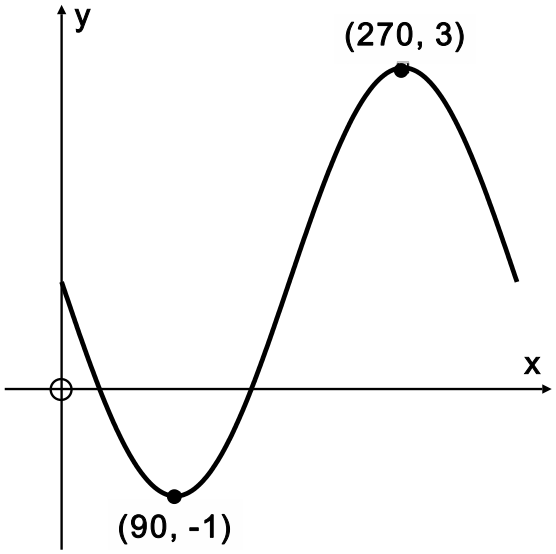
4. $y = a\cos x + q$



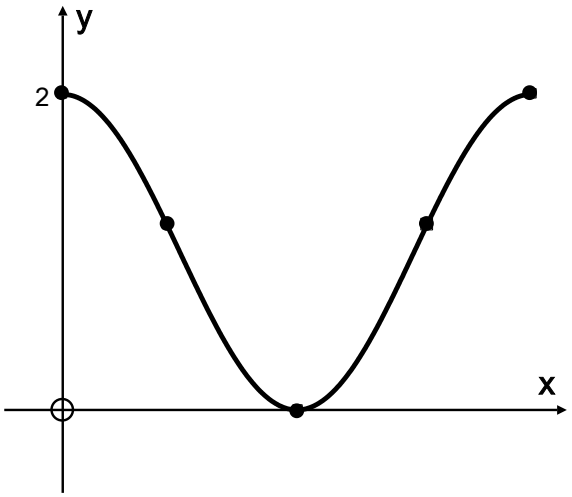
5. $y = a\cos x + q$



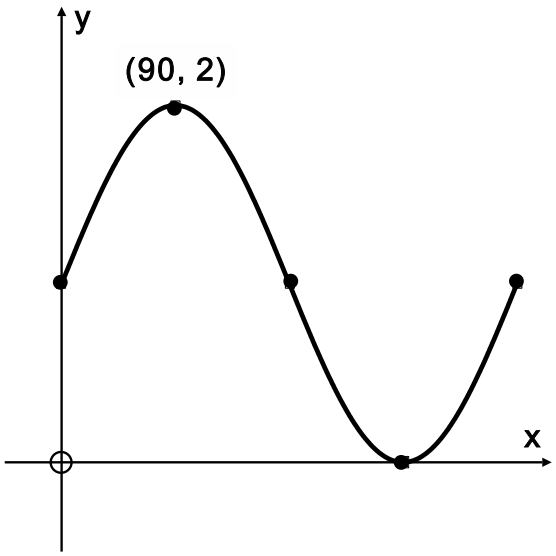
6. $y = a\sin x + q$



7. $y = a\cos x + q$



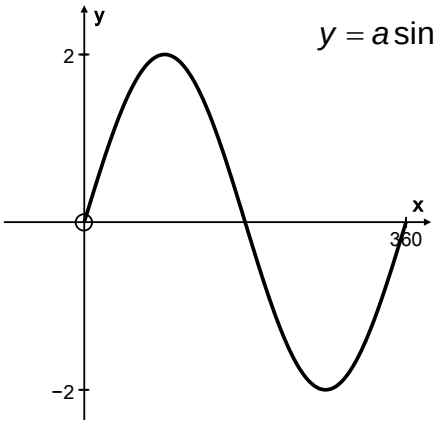
8. $y = a\sin x + q$



Exercise B:

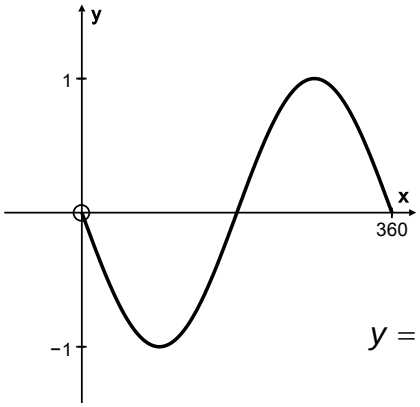
Find the values of *a* and *q* in each of the following:

1.



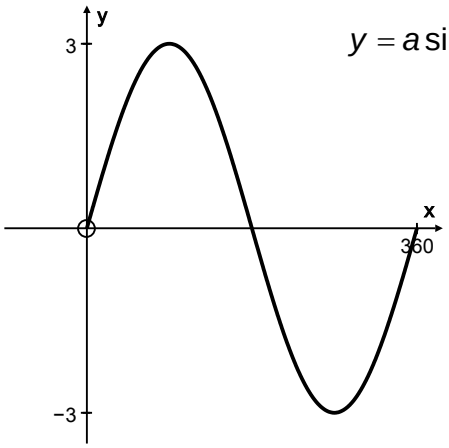
$y = a\sin x + q$

2.



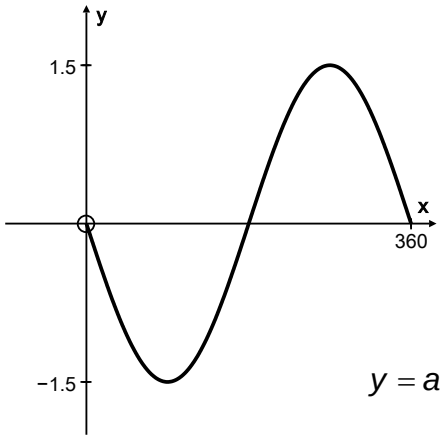
$y = a\sin x + q$

3.



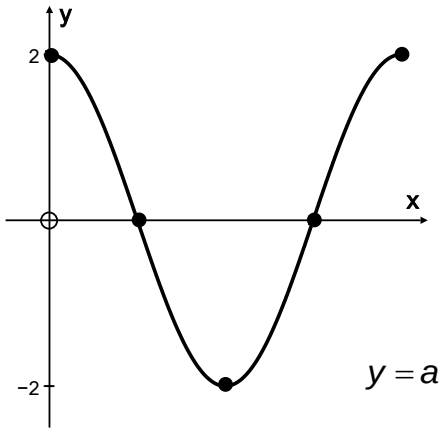
$y = a\sin x + q$

4.



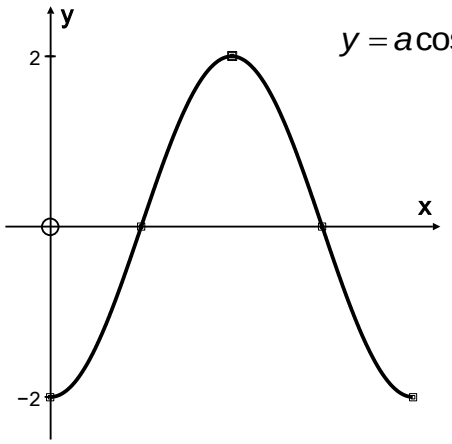
$y = a\sin x + q$

5.



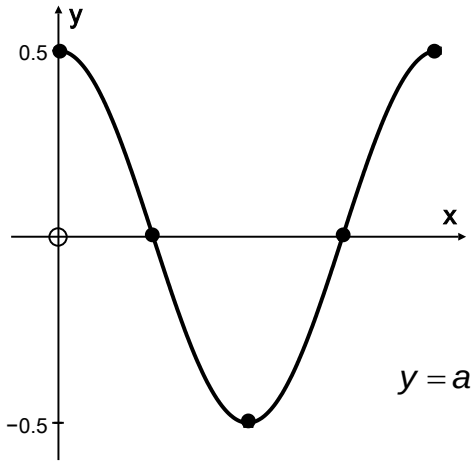
$y = a\cos x + q$

6.



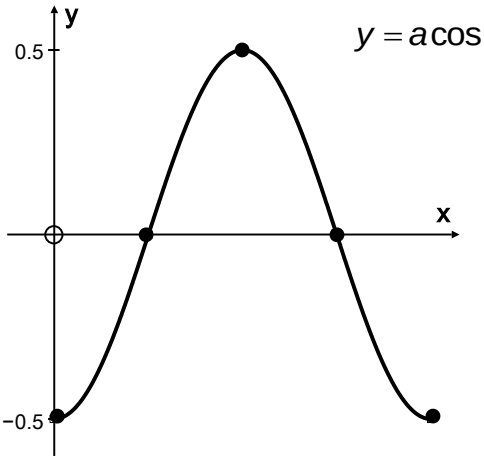
$y = a\cos x + q$

7.



$y = a\cos x + q$

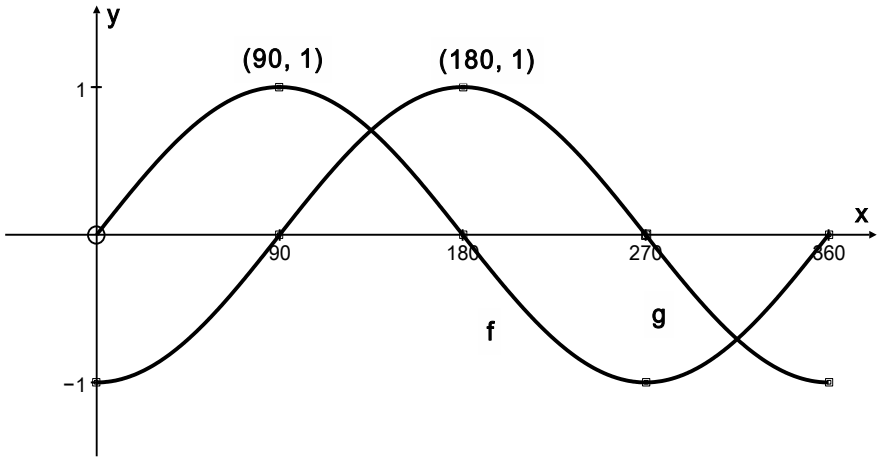
8.



$y = a\cos x + q$

TRIG GRAPHS WORKSHEET (1)

1. Given: $f(x) = a\sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ; 360^\circ]$



Period of f:

Period of g:

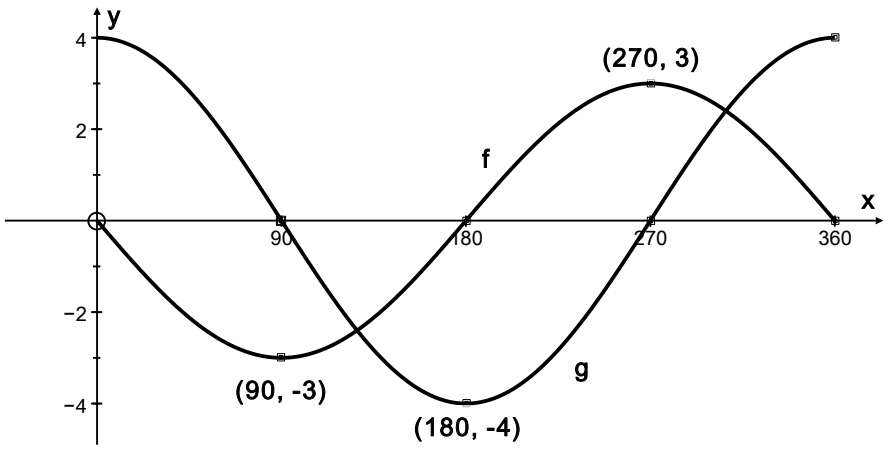
Amplitude of f:

Amplitude of g:

a) Find the values of a and b

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ; 360^\circ]$

2. Given: $f(x) = a\sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ; 360^\circ]$



Period of f:

Period of g:

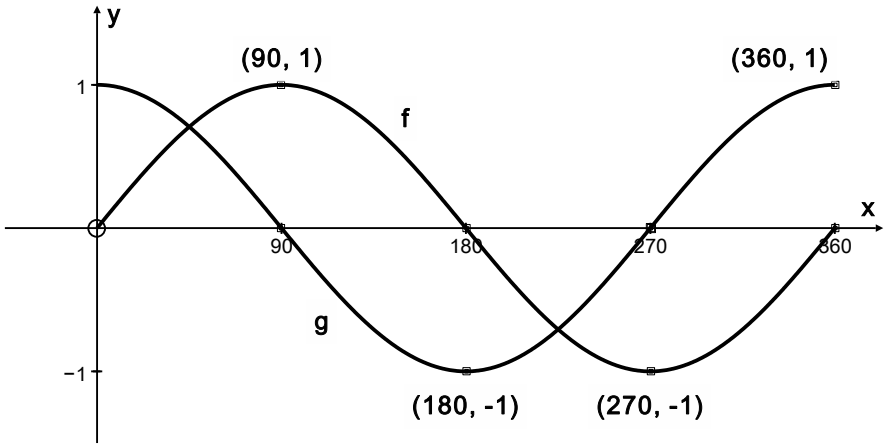
Amplitude of f:

Amplitude of g:

a) Find the values of a and b

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ; 360^\circ]$

3. Given: $f(x) = a \sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ; 360^\circ]$



Period of f:

Period of g:

Amplitude of f:

Amplitude of g:

a) Find the values of a and b

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ; 360^\circ]$

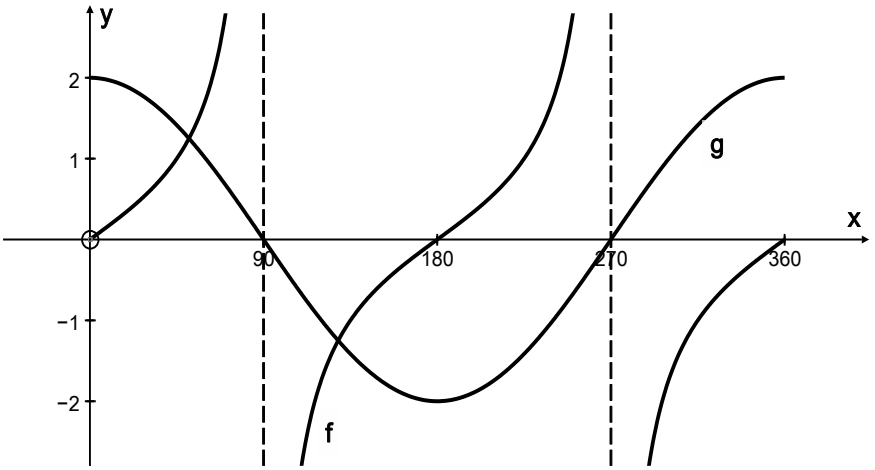
TRIG GRAPHS WORKSHEET (2)

1. The following graphs have been drawn below:

$f(x) = \tan x$ *and* $g(x) = a\cos x$

a) Determine the value of a.

b) If f(x) is shifted 2 units upwards, it represents the function h. State the <u>defining equation</u> of h.



c) Write down the values of x for which g(x) – f(x) = 2

d) Write down the <u>range</u> of g.

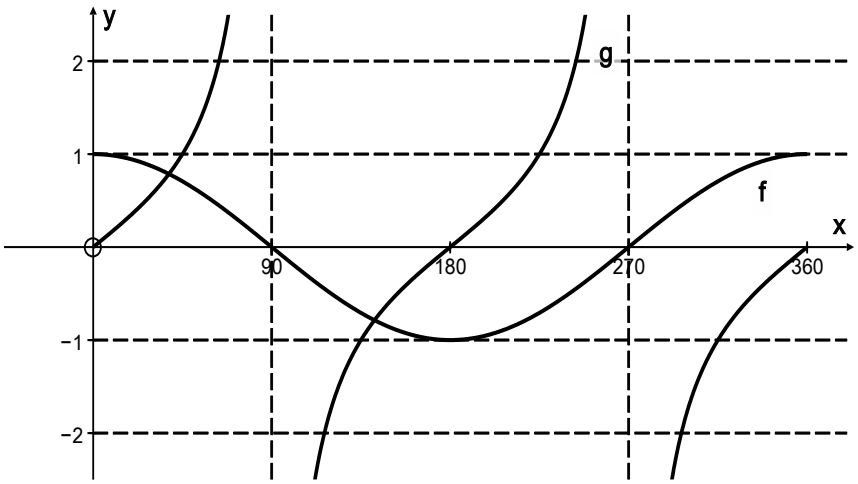
e) Write down the <u>period</u> of f.

2. The diagram below represents the graphs of f(x) and g(x) = tan θ

a) Give the <u>equation</u> of f(x)

b) What is the period of g(x)?

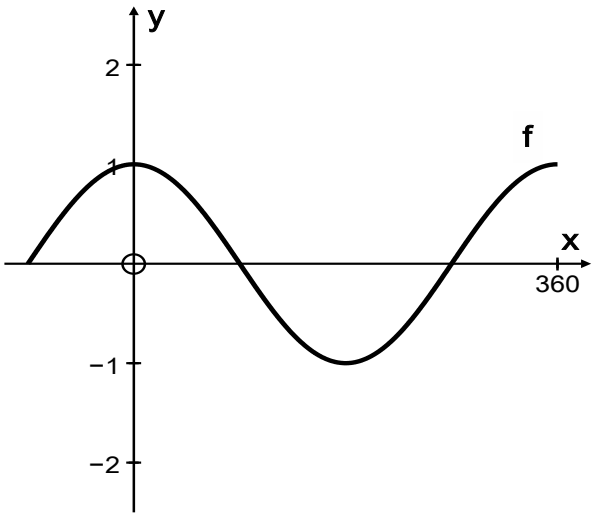
c) Write down the equations of the two lines that are the <u>asymptotes</u> of g(x) = tan



d) What is the <u>range</u> of g.

e) For what values of x is f(x) – g(x) = 1?

3. The diagram alongside shows the graph of f



a) What is the <u>equation of f</u> ?

f(x) =

b) Write down the <u>amplitude and period</u>

of f

Amplitude: _____

Period: _____

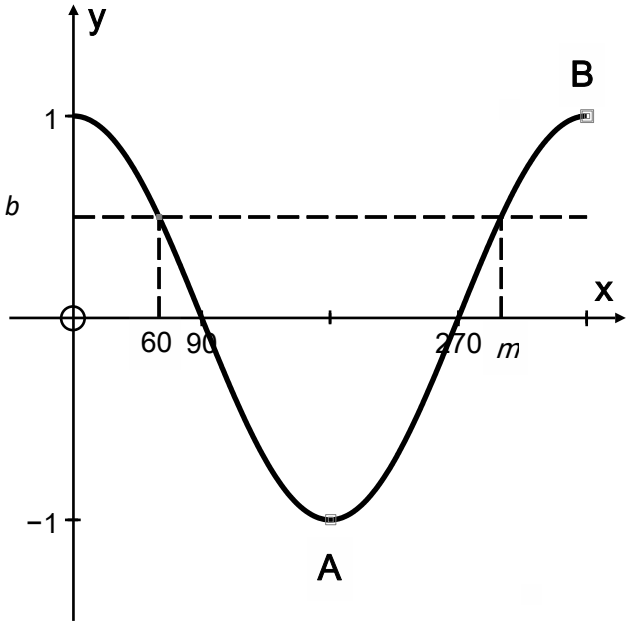
c) On the same set of axes <u>draw the graph</u> of g(x) = sinx – 1. Show clearly all the intercepts and turning points of both curves.

d) Label the <u>points of intersection</u> of both graphs f and g , using capital letters A, B, ...

e) Write down the <u>solution of the equation</u> cos x = sinx – 1 for $x \in [0^\circ ; 360^\circ]$

TRIG GRAPHS WORKSHEET (3)

1. The sketch $f(x)=\cos x$ is given



1.1. Determine the co-ordinates of A and B

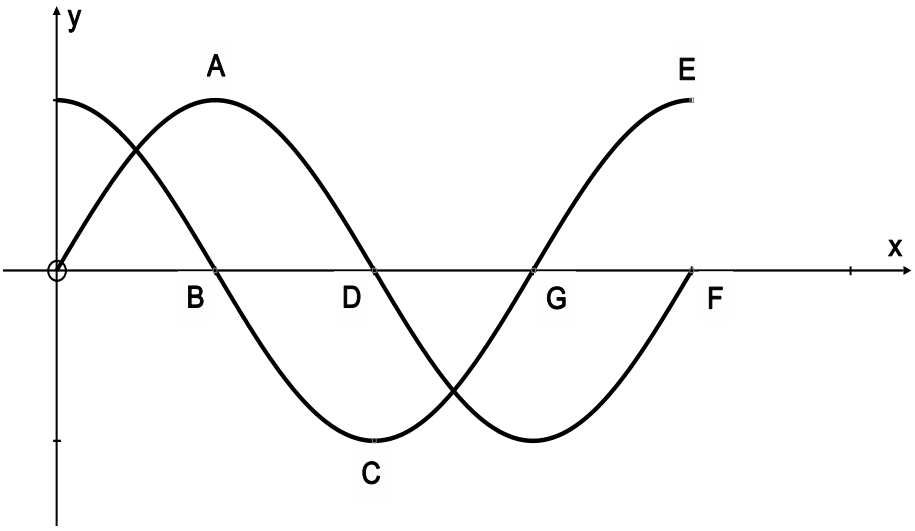
1.2. What is the value of m?

1.3. Find the value of *b* (use a calculator)

1.4. Use the graph to solve $\cos x + 1 = 0$

2. The figure represents:

$h(x)=\sin\theta$ *and* $g(x)=\cos\theta$



2.1. Give the co-ordinates of:

A _____

B _____

C _____

D _____

E _____

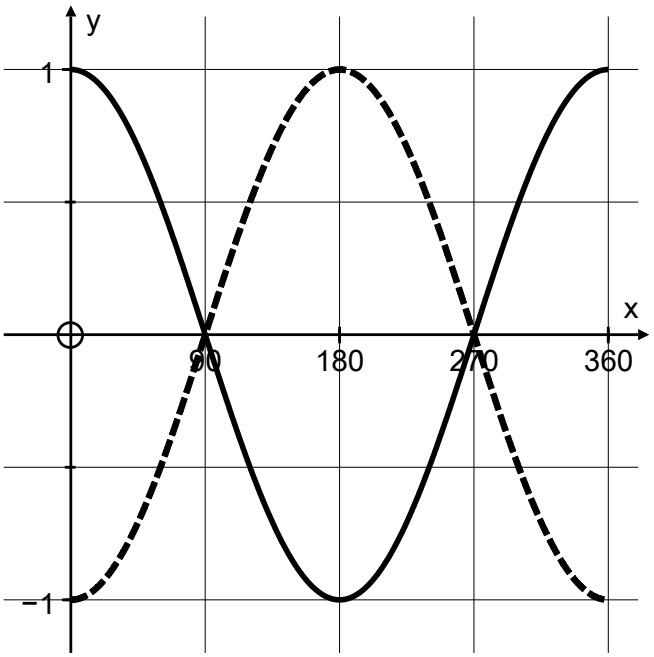
F _____

2.2. What is the minimum value of h?

2.3. For which values of θ is $\sin\theta.\cos\theta < 0$?

3. The graph of $y = \cos x,\ 0^\circ \le x \le 360^\circ$ is shown.

Some transformations were performed to produce the dotted curve.

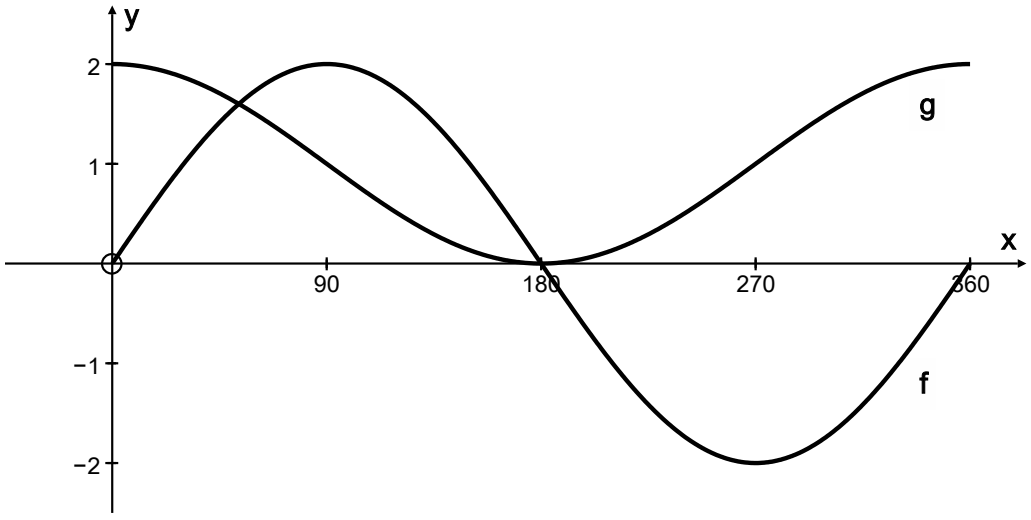


3.1. What is the period of $y = \cos x$?

3.2. Write down the equation that describes the dotted line.

3.3. If the dotted line is moved 2 units downwards, write down the equation for the new curve.

4. The diagram below represents the graphs of the functions $f(x) = a\sin x$ *and* $g(x) = \cos x + q$



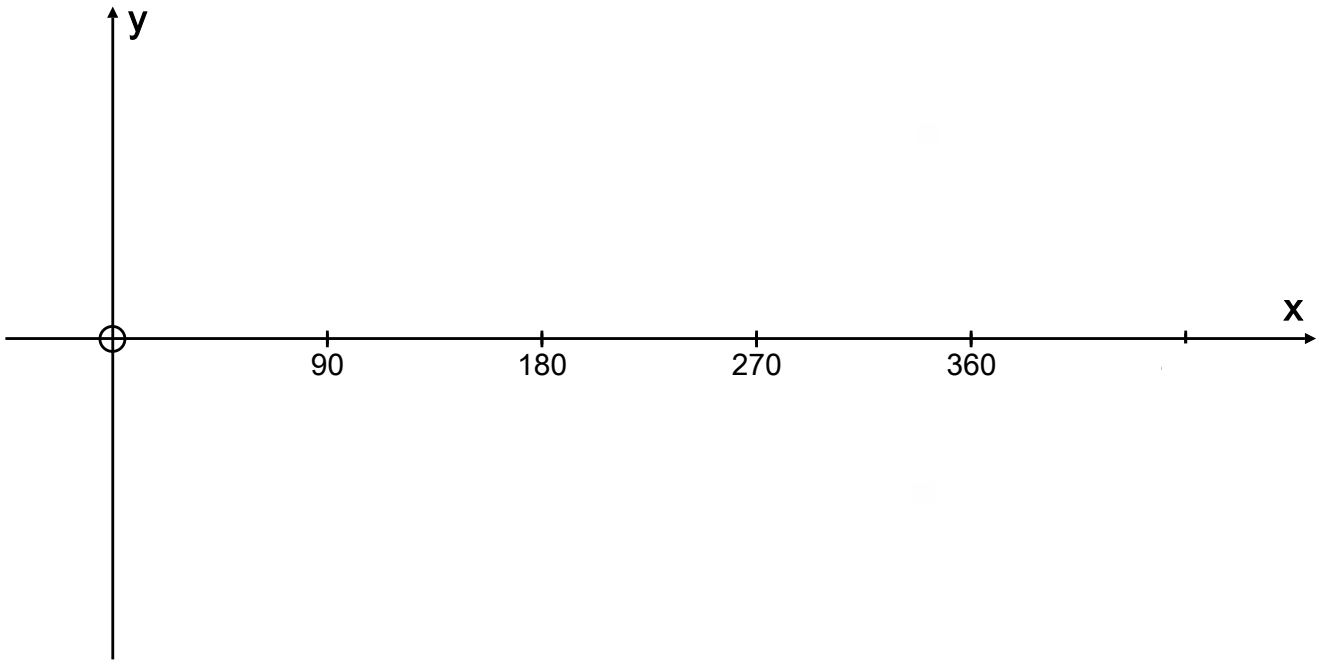
4.1. What is the value of a?

4.2. Write down the value of q

4.3. What is the range of g?

4.4. What is the value of f(270°)?

5. Draw a graph of $y = \cos x$ *for* $x \in [0^\circ ; 360^\circ]$



5.1. What is the amplitude of the graph?

5.2. What is the period of the graph?

5.3. For what values of x is $\cos x = \frac{1}{2}$?

5.4. For what values of x is $\cos x \leq 0$?

5.5. What does the equation of $y = \cos x$ become if the graph is shifted 3 units down?